CVICU TRAINER

CVICU Badge Buddies

How to make them: print at **100% / Actual size** (not "fit to page"), ideally on card stock. Cut on the solid lines, **fold on the dashed line** so the front and back face out, then laminate or slip into a badge sleeve. Each card is ID-badge size (3.375 × 2.125 in).

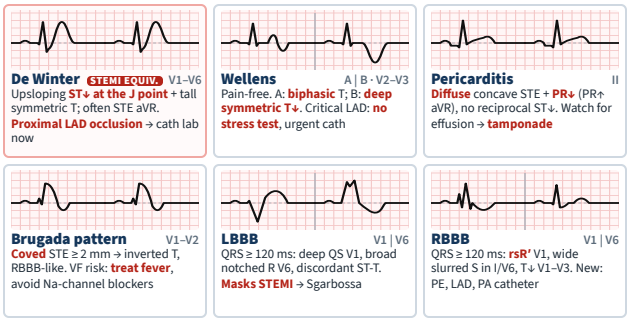
CVICU BADGE BUDDY · 12-lead: walls & arteries · 1 · front

I HIGH LATERAL LCx · diagonal (LAD)	**aVR** LEFT MAIN / 3-VESSEL STE aVR + diffuse ST↓	**V1** SEPTAL LAD (septal)	**V4** ANTERIOR LAD
II INFERIOR RCA (~80%) or LCx	**aVL** HIGH LATERAL LCx · diagonal (LAD)	**V2** SEPTAL LAD (septal)	**V5** LOW LATERAL LCx · LAD
III INFERIOR RCA	**aVF** INFERIOR RCA or LCx	**V3** ANTERIOR LAD	**V6** LOW LATERAL LCx

↔ **Reciprocal ST↓ supports STEMI:** inferior STE ↔ ST↓ in I/aVL · lateral STE ↔ ST↓ in II, III, aVF

\+ **Posterior:** ST↓ + tall R in V1–V3 → record V7–V9 (LCx/RCA). **RV infarct:** inferior STEMI + low BP → V4R STE (proximal RCA): fluids, avoid nitrates

CVICU BADGE BUDDY · STEMI equivalents, mimics & BBBs · 1 · back



De Winter STEMI EQUIV. V1–V6
Upsloping **ST↓ at the J point** + tall symmetric T; often STE aVR. **Proximal LAD occlusion** → cath lab now

Wellens A | B · V2–V3
Pain-free. A: **biphasic** T; B: **deep symmetric T↓**. Critical LAD: **no stress test**, urgent cath

Pericarditis II
Diffuse concave STE + **PR↓** (PR↑ aVR), no reciprocal ST↓. Watch for effusion → **tamponade**

Brugada pattern V1–V2
Coved STE ≥ 2 mm → inverted T, RBBB-like. VF risk: **treat fever**, avoid Na-channel blockers

LBBB V1 | V6
QRS ≥ 120 ms: deep QS V1, broad notched R V6, discordant ST-T. **Masks STEMI** → Sgarbossa

RBBB V1 | V6
QRS ≥ 120 ms: **rsR′** V1, wide slurred S in I/V6, T↓ V1–V3. New: PE, LAD, PA catheter

Sgarbossa (LBBB / paced): concordant STE ≥ 1 mm or ST↓ ≥ 1 mm in V1–V3 = acute MI. **Bifascicular** (RBBB + LAFB/LPFB) + 1st-degree block → pacing ready.

CVICU BADGE BUDDY · Swan-Ganz & hemodynamics · 2 · front



RA / CVP	RV	PA	PAWP ≈ LA	LV
5	**25/5**	**25/10**	**10**	**120/8**
nickel	quarter / nickel	quarter / dime	dime	dollar
2–6 mmHg	15–30 / 0–8	15–30 / 4–12	6–12 mmHg	100–140 / 3–12
15–25 cm	25–35 cm	40–50 cm	45–55 cm	aorta 100–140/60–90

"**Nickel** in the atrium, **quarter over nickel** in the RV, **quarter over dime** in the PA, **dime** for the wedge, a **dollar** in the LV." Depths: from the right IJ. PA mean 9–18. **PAD ≈ PAWP** (gap > 5 → high PVR).

MAP	70–105 mmHg	**SVR**	800–1200 dyn·s/cm^5
CO	4–8 L/min	**PVR**	< 250 dyn·s/cm^5
CI	2.5–4 L/min/m^2	**SvO_2 · $ScvO_2$**	60–80% · ≥ 70%
SV	60–100 mL/beat	**Lactate**	< 2 mmol/L

CVICU BADGE BUDDY · Shock profiles · 2 · back

	CVP	PAWP	CI	SVR	SvO_2
Hypovolemic / bleeding	↓	↓	↓	↑	↓
Cardiogenic (LV)	↑	↑	↓	↑	↓
RV failure	↑	N / ↓	↓	↑	↓
Tamponade	↑	↑ (≈ CVP)	↓	↑	↓
Massive PE	↑	N / ↓	↓	↑	↓
Distributive (sepsis, vasoplegia)	↓ / N	↓ / N	↑ / N	↓	↑ / N

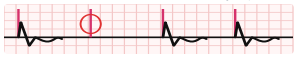
After cardiac surgery: CVP ≈ PAD ≈ PAWP + falling CI + chest tube output that suddenly stops → **tamponade**: notify the surgeon now. High CI + low SVR soon after bypass → vasoplegia.

Cold & wet (low CI, high PAWP) → inotrope + diuretic · **Warm** (high CI) with low SVR → vasopressor · **Low CVP + PAWP** → volume or blood. Follow orders and unit protocols.

CVICU BADGE BUDDY · Temporary pacing · VVI troubleshooting · 3 · front

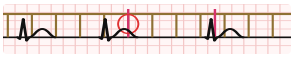
V Paces the ventricle · V Senses the ventricle · I Inhibits when it senses a beat

LOSS OF CAPTURE — spike, no QRS



Patient first · ↑ output · check battery & every connection · turn onto **left side** · notify

UNDERSENSING — fence too HIGH



QRS can't clear the fence → spikes on QRS/T (R-on-T). mV **down** · document · notify MD

OVERSENSING — fence too LOW



T waves/noise clear the fence → pauses. Remove cause · mV **up** · document · notify MD

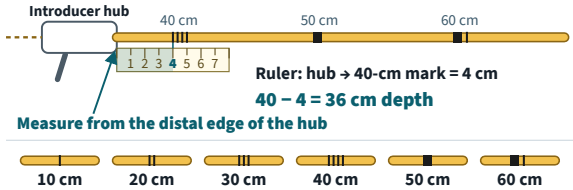
NO OUTPUT — no spikes at all



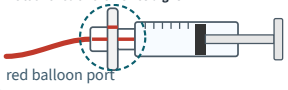
Power, connections, new battery or pacer · **ZOLL pads / TCP** if no rhythm

Notify: loss of capture · sudden ↑ threshold · failure to sense despite troubleshooting · instability · site bleeding. **Unstable?** Adjust settings, then notify the MD.

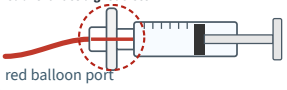
CVICU BADGE BUDDY · Transvenous: depth & balloon lock · 3 · back



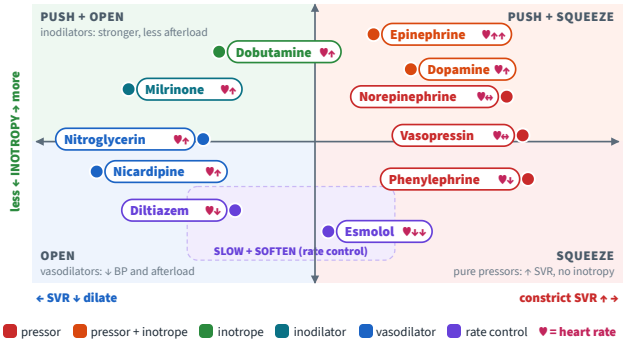
Locked: correct
The **lock's red line is NOT straight**



Unlocked: lock it now
Red line runs **straight across**



CVICU BADGE BUDDY · Vasoactive drug map · 4 · front



pressor · pressor + inotrope · inotrope · inodilator · vasodilator · rate control · ♥ = heart rate

CVICU BADGE BUDDY · Vasoactives: what they do · 4 · back

Drug	Acts on	HR	Ino	SV	SVR	Use it for
Norepinephrine	α1 ≫ β1	↔	↑	↔/↑	↑↑	First-line: septic shock, vasoplegia
Vasopressin	V1	↔	↔	↔	↑↑	2nd pressor; spares catechols; RV-friendly
Phenylephrine	α1	↓ reflex	↔	↓/↔	↑↑	Low SVR + tachycardia; SAM / LVOT
Epinephrine	β1 β2 (α high)	↑↑	↑↑	↑	↓→↑	Low output, refractory shock, arrest
Dopamine	DA → β1 → α1	↑	↑	↑	↑ high	Bradycardic hypotension (2nd line)
Dobutamine	β1 (β2)	↑	↑↑	↑↑	↓	Low output with an OK MAP
Milrinone	PDE-3	↑ slight	↑	↑	↓↓	Low output, RV failure, on β-blockers
Nitroglycerin	NO (venous)	↑ reflex	↔	↓/↔	↓	Ischemia, pulmonary edema, post-CABG HTN
Nicardipine	Ca^{2+} (arterial)	↑ reflex	↔	↔/↑	↓↓	Strict BP control (after HR control)
Esmolol	β1 block	↓↓	↓	↔/↓	↔	Dissection anti-impulse, SVT/AF rate
Diltiazem	Ca^{2+} (node)	↓	↓	↔/↓	↓	AF/flutter rate (avoid if EF low)

SV and CO also depend on volume and LV function: more SVR can lower SV in a failing LV. Pulmonary vasodilators (inhaled NO, epoprostenol) ↓ PVR for the RV. Doses: your unit's protocol.

Educational reference. Follow your unit's protocols, provider orders and device instructions. © 2026 Alexander Bohn · CVICU Trainer